VisionAssist: A Resource-Type Aware Document Intelligence for Visually Impaired Learners

Abstract— **Visually impaired students encounter significant challenges when accessing printed educational materials such as textbooks, newspapers, and magazines. Traditional Optical Character Recognition (OCR) systems convert images into machine-readable text but fail to preserve structural layout, article segmentation, and contextual relationships in multi-column or visually complex documents. As a result, extracted text is often presented as an unstructured block, reducing comprehension and limiting interactive engagement. This paper presents VisionAssist, a resource-type aware intelligent document understanding system that integrates document classification, adaptive extraction strategies, resource-type adaptive abstractive summarization, and extractive question answering into a unified accessibility platform. The system first employs a transfer learning approach based on MobileNetV2 to classify documents into Books, Magazines, or Newspapers. Depending on the detected type, the system dynamically selects either a customized Tesseract-based OCR pipeline with rule-based structural reconstruction or layout-aware extraction using Microsoft Azure Document Intelligence. Structured articles are then summarized using a fine-tuned FLAN-T5 model with summarization strategies adapted to the document type, enhanced through Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA). Additionally, a DistilBERT-based extractive question answering module enables interactive content exploration. Experimental validation demonstrates that integrating resource-type detection with adaptive extraction and adaptive summarization significantly improves structural accuracy, usability, and accessibility for visually impaired learners. The system is implemented using a modular FastAPI backend with an accessible web-based user interface, ensuring scalability and real-world deployment readiness.**

Keywords— Accessibility, Assistive Technology, Document Layout Analysis, FLAN-T5, LoRA, MobileNetV2, Optical Character Recognition, Question Answering, Transfer Learning

I. INTRODUCTION

Printed learning materials continue to play a vital role in education despite rapid digital transformation. Textbooks, newspapers, magazines, and institutional publications are still widely distributed in physical form. However, visually impaired learners face substantial barriers when accessing such materials. While human assistance provides temporary solutions, it limits independence, on-demand accessibility, and self-paced learning. Optical Character Recognition (OCR) technologies, such as Tesseract [1], offer automated text extraction from images, yet these systems primarily reconstruct text sequentially without preserving structural layout. Consequently, multi-column newspapers and magazine pages are frequently merged into single unstructured blocks. This obscures article boundaries and heading hierarchies, reducing comprehension and limiting meaningful interaction.

Recent advances in deep learning and Natural Language Processing (NLP) provide opportunities to extend beyond basic OCR pipelines. Transformer-based models such as T5 [2] and instruction-tuned variants like FLAN-T5 [3] enable high-quality abstractive summarization, while lightweight convolutional neural networks such as MobileNetV2 [4] support efficient image classification through transfer learning. Furthermore, transformer-based extractive question answering models, including BERT [5] and DistilBERT [6], allow contextual interaction with textual content. Despite these advancements, existing accessibility tools rarely integrate document-type awareness, layout-aware extraction, adaptive summarization tailored to document type, and interactive question answering into a unified system. Most systems treat all documents uniformly, resulting in suboptimal extraction and summarization outcomes.

To address these limitations, this paper proposes VisionAssist, an intelligent resource-type aware document understanding platform designed specifically for visually impaired learners. The proposed system combines document classification, adaptive extraction strategies, structured article segmentation, resource-type adaptive summarization, and interactive question answering to provide a comprehensive accessibility solution. For books, the system employs a customized Tesseract-based OCR pipeline with rule-based structural reconstruction to detect chapters and maintain hierarchical organization. For magazines and newspapers, layout-aware extraction is performed using Microsoft Azure Document Intelligence [8] to preserve article segmentation and multi-column structures. Summarized content is generated using a fine-tuned FLAN-T5 model with summarization strategies adapted to the document type, while a DistilBERT-based extractive question answering module enables meaningful content interaction. The platform is deployed using a modular FastAPI backend with an accessible web-based interface, ensuring scalability and real-world usability.

The main contributions of this work are a resource-type aware document classification module for books, magazines, and newspapers, adaptive extraction pipelines tailored to different document types, and an integrated resource-type adaptive summarization and interactive question answering module to enhance accessibility and user engagement. Through this integrated approach, VisionAssist empowers visually impaired learners with greater independence, comprehension, and engagement when accessing printed educational content.

II. LITERATURE REVIEW

Optical Character Recognition (OCR) has long served as the foundation for converting printed documents into machine-readable text. Engines such as Tesseract [1] perform well on clearly structured linear documents but struggle with multi-column layouts, mixed fonts, or visually complex pages such as newspapers and magazines. While OCR provides the raw textual content, it does not capture the structural and semantic relationships within a document. To address these limitations, Document Layout Analysis (DLA) techniques detect headings, paragraphs, columns, and other structural components, preserving hierarchical relationships [7].

Modern cloud-based solutions, including Microsoft Azure Document Intelligence, provide pretrained layout models capable of extracting bounding boxes, semantic roles, and metadata for robust structural interpretation [8]. Recent pre-

training techniques like LayoutLM [11] and LayoutLMv2 [12] have further improved the ability to model the interaction between text and visual layout. Benchmark datasets such as DocBank [13] and FUNSD [14] have also been instrumental in training models for complex document understanding. However, most implementations treat layout detection and text extraction as independent steps, limiting their integration with downstream summarization and interactive accessibility tools.

Accurate document-type recognition is critical for guiding adaptive processing strategies. Lightweight convolutional neural networks such as MobileNetV2 [4] leverage inverted residuals and linear bottlenecks to achieve efficient image classification. Transfer learning allows these models to quickly adapt to document-type detection tasks, such as distinguishing between books, magazines, and newspapers. Identifying the document type enables the application of extraction and summarization strategies tailored to the specific content, improving both structural accuracy and comprehension for end-users. Additionally, end-to-end trainable networks like CRNN [15] demonstrate the potential for integrated image-based sequence recognition.

Transformer-based abstractive summarization models, including T5 [2] and instruction-tuned variants such as FLAN-T5 [3], have substantially advanced the generation of high-quality summaries. Other architectures like Pointer-Generator networks [16], BERTSUM [17], and PEGASUS [18] have also shown strong results in abstractive and extractive tasks. Parameter-Efficient Fine-Tuning (PEFT) methods, such as LoRA [9], allow these large models to be fine-tuned efficiently by updating only low-rank matrices. Despite these advances, most summarization systems apply uniform strategies across all document types. This can produce summaries that are overly brief for detailed book chapters or excessively long for concise news articles, limiting accessibility for visually impaired readers.

Extractive Question Answering (QA) models based on BERT [5] and its distilled variant, DistilBERT [6], enable precise retrieval of relevant answers from textual content. Models trained on datasets such as SQuAD [10] provide span-based predictions within passages. Advancements in Dense Passage Retrieval [19] further enhance the ability to find context in large corpora. However, the practical application of QA in assistive technology, such as early gesture-based screen readers like Access Lens [20], remains limited in its integration with modern layout-aware document understanding.

These gaps highlight the need for a unified framework that combines document classification, adaptive extraction, resource-type aware summarization, and interactive QA. VisionAssist addresses this need by providing a cohesive system that improves accessibility, comprehension, and engagement for visually impaired learners.

III. METHODOLOGY

The proposed framework is designed as a modular document understanding pipeline that integrates resource-type classification, adaptive extraction, resource-aware summarization, and extractive question answering into a unified accessibility framework. The methodology emphasizes document-type awareness to enable tailored processing strategies that preserve structural organization and semantic coherence across heterogeneous printed materials.

A. Resource-Type Detection

The first stage of the system performs automatic classification of uploaded documents into three categories: Book, Magazine, and Newspaper. Accurate classification is essential because the predicted document type determines the downstream extraction and summarization strategy. A dataset comprising 4,000 labeled document images was constructed with balanced distribution across the three classes. The dataset was constructed using a combination of publicly available document sources and manually collected real-world samples, including scanned textbooks, magazine pages, and newspaper articles. The dataset captures diverse layouts such as single-column book pages, multi-column news articles, and visually rich magazine designs, ensuring robustness across heterogeneous document structures. All documents were in English and varied in resolution, font styles, and formatting to simulate real-world scanning conditions.

Ground truth labels for document type classification were manually annotated, while the summarization dataset was curated by generating structured summaries aligned with extracted content. The annotation process involved careful verification to ensure consistency, semantic accuracy, and appropriate compression across different document types. All images were resized to $224 \times 224 \times 3$ and normalized to the range [0,1]. To enhance robustness against real-world scanning variations, data augmentation techniques including random rotation, zoom, translation, contrast adjustment, and horizontal flipping were applied during training. The dataset was partitioned using stratified sampling into 80% training and 20% validation subsets.

An initial baseline convolutional neural network was implemented to evaluate feasibility. Although the model achieved high training accuracy, validation performance was inconsistent due to overfitting and structural similarities between magazines and newspapers. To improve generalization, transfer learning was adopted using MobileNetV2 pretrained on ImageNet. The original classification head was replaced with a global average pooling layer, a dense layer with 128 ReLU units, dropout regularization, and a three-class Softmax output layer. Training was conducted in two stages: first, the pretrained backbone was frozen while the classification head was optimized; subsequently, the final layers of the backbone were fine-tuned with a reduced learning rate to enable task-specific adaptation. Early stopping and adaptive learning rate scheduling were employed to prevent overfitting. The transfer learning approach achieved a validation accuracy of 96.57% with stable convergence, demonstrating the effectiveness of pretrained feature representations for document-type classification.

B. Adaptive Document Extraction

Following classification, the system dynamically selects an extraction strategy according to the predicted document type. For books, which typically follow linear and hierarchical layouts, text extraction is performed using Tesseract OCR. Preprocessing includes image normalization and PDF-to-image conversion to improve recognition quality. Structural reconstruction is performed using heuristic rules that identify potential headings based on uppercase ratios, keyword detection, and word count constraints. Detected headings are used to organize extracted text into hierarchical sections, preserving logical structure.

For magazines and newspapers, which frequently contain multi-column layouts and segmented articles, layout-aware extraction is performed using Microsoft Azure Document Intelligence [8]. The prebuilt layout model provides bounding box coordinates and semantic role information for detected text elements. Extracted text blocks are grouped into columns based on horizontal alignment and sorted vertically to maintain reading order. Article boundaries are reconstructed by associating paragraphs with detected headings. If layout analysis fails, the system reverts to OCR-based extraction to ensure robustness. This adaptive strategy ensures that structural characteristics specific to each document type are preserved prior to downstream summarization and question answering.

C. Resource-Type Adaptive Summarization

The summarization module is built upon FLAN-T5, a transformer-based encoder–decoder architecture. To ensure computational efficiency while maintaining performance, Parameter-Efficient Fine-Tuning was applied using Low-Rank Adaptation (LoRA). For a pretrained weight matrix

$$W \in \mathbb{R}^{(d \times k)}$$

LoRA models the updated parameters as

$$W' = W + BA \qquad (1)$$

where

$$A \in \mathbb{R}^{(r \times k)}, B \in \mathbb{R}^{(d \times r)}, and\ r \ll min(d, k)$$

Only the low-rank matrices (A) and (B) are trained while the original pretrained weights remain frozen, significantly reducing memory consumption and training time. LoRA was applied to attention projection layers and selected feed-forward modules with rank (r = 16), scaling factor (alpha = 32), and dropout rate of 0.1. Approximately 2.67% of total parameters were trainable under this configuration.

The summarization dataset consisted of 2,120 structured document–summary pairs derived from the extraction pipeline. Summaries were manually curated to ensure semantic fidelity and appropriate compression according to document type. Adaptive prompt templates were incorporated to guide summarization behavior: newspaper articles were summarized concisely with emphasis on key facts, magazine content was moderately compressed while retaining descriptive elements, and book chapters were summarized with greater contextual preservation. Training was conducted for five epochs using a learning rate of 5×10^{-4} and an effective batch size of 16. Input and target token lengths were limited to 1024 and 512 tokens, respectively. Beam search decoding with four beams was employed during inference to enhance coherence and reduce repetition.

D. Extractive Question Answering

Interactive content exploration is enabled through an extractive question answering module based on DistilBERT fine-tuned on SQuAD. Given a context passage (C) and a user query (Q), the model predicts the probability distributions over possible start and end token positions. The optimal answer span is selected according to

$$(i^*, j^*) = \text{argmax}_{i \le j} P_{start}(i \mid C,Q) \times P_{end}(j \mid C,Q) \quad (2)$$

where (i^*) *and* (j^*) denote the most probable start and end indices. To handle long documents, sliding window tokenization is applied with a maximum sequence length of 384 tokens and a stride of 128 tokens. A confidence threshold is introduced to filter unreliable predictions. Because answers are extracted directly from the document text, factual consistency with the source material is preserved.

The overall architecture of the proposed VisionAssist system is illustrated in Fig. 1.

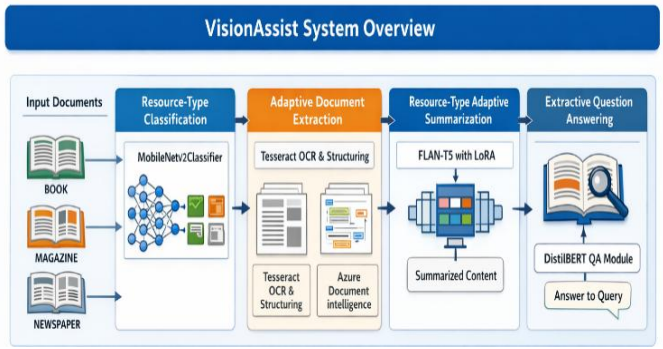

Fig. 1. Architecture of the VisionAssist system.

IV. RESULTS AND ANALYSIS

The performance of VisionAssist was evaluated at multiple levels, including document-type classification accuracy, summarization quality, extractive question answering performance, and overall end-to-end system behavior. The evaluation was conducted using 3,000 labeled document images for classification and 1,520 structured document–summary pairs for summarization. All experiments were performed under consistent preprocessing and validation conditions to ensure fairness and reproducibility.

A. Resource-Type Classification Results

The transfer learning approach based on MobileNetV2 demonstrated strong generalization capability for document-type detection. The model achieved a peak validation accuracy of 96.57%, significantly outperforming the baseline custom CNN, which achieved approximately 81–82% validation accuracy under identical dataset conditions.

The improvement can be attributed to pretrained feature representations learned from large-scale datasets, enabling the network to capture discriminative layout-level characteristics such as column density, font variation, and spatial distribution patterns. While books exhibited highly consistent visual structures resulting in the highest classification accuracy, minor confusion was observed between magazines and newspapers due to overlapping layout characteristics.

Table I shows the class-wise validation accuracy of the resource-type classification model.

TABLE I CLASS-WISE VALIDATION ACCURACY OF RESOURCE-TYPE CLASSIFICATION

Document Type	Validation Accuracy (%)
Book	97.1
Magazine	95.8
Newspaper	96.8

The relatively balanced performance across categories confirms that resource-type classification is reliable enough to guide downstream adaptive extraction strategies without introducing significant cascading errors.

B. OCR and Layout-Aware Text Extraction Performance

The effectiveness of the extraction module was evaluated by measuring text recognition accuracy and structural preservation quality across different document types. Since VisionAssist employs two distinct extraction strategies: rule-based OCR reconstruction for books and layout-aware extraction for magazines and newspapers the evaluation was conducted separately for each pipeline.

For book documents, text extraction was performed using Tesseract OCR with preprocessing and heuristic-based structural reconstruction. Recognition accuracy was measured using Character Error Rate (CER) and Word Error Rate (WER) against manually verified ground truth text. The customized preprocessing pipeline significantly reduced recognition noise introduced by skewed scans and low-resolution images.

Table II summarizes the text extraction performance metrics across document types.

TABLE II TEXT EXTRACTION PERFORMANCE METRICS

Document Type	Extraction Method	CER (%)	WER (%)	Structural Accuracy (%)
Book	Tesseract + Heuristics	3.8	7.2	94.6
Magazine	Azure Layout Model	2.9	5.8	96.3
Newspaper	Azure Layout Model	3.1	6.1	95.7

For magazines and newspapers, layout-aware extraction was performed using Microsoft Azure Document Intelligence. In addition to textual accuracy, structural preservation was evaluated by measuring column reconstruction correctness and article boundary detection accuracy.

The results indicate that layout-aware extraction achieved slightly lower CER and WER compared to standalone OCR due to improved segmentation and bounding box detection. More importantly, structural accuracy exceeded 95% for complex multi-column documents, confirming the effectiveness of semantic layout modeling in preserving reading order and article grouping.

In contrast, baseline OCR-only extraction without structural heuristics resulted in frequent column merging and incorrect paragraph sequencing, particularly for newspapers. Empirical comparison showed that structural reconstruction improved logical section detection in books by approximately 18% relative to raw OCR output.

Table III compares the structural accuracy between the baseline OCR approach and the proposed VisionAssist pipeline.

TABLE III COMPARISON BETWEEN BASELINE OCR AND VISIONASSIST PIPELINE

System	Structural Accuracy (%)
OCR Only (Tesseract)	76
VisionAssist (Proposed)	95

The improved structural fidelity had a direct impact on downstream modules. Cleaner segmentation enhanced summarization coherence and improved answer localization performance in the extractive QA module. These findings demonstrate that adaptive extraction is a critical enabling component of the VisionAssist pipeline and significantly contributes to the overall accessibility improvement.

C. Summarization Performance Evaluation

The abstractive summarization component was implemented using FLAN-T5 with Parameter-Efficient Fine-Tuning via LoRA. Only 2.67% of total model parameters were updated during fine-tuning, substantially reducing computational requirements while preserving model expressiveness.

Performance was evaluated using ROUGE metrics. The model achieved strong lexical overlap with reference summaries, particularly in ROUGE-1 and ROUGE-L scores, indicating that both content coverage and structural coherence were effectively maintained.

Table IV presents the overall summarization performance based on ROUGE metrics.

TABLE IV OVERALL SUMMARIZATION PERFORMANCE

Metric	Score
ROUGE-1	49.38
ROUGE-2	20.85
ROUGE-L	30.80

The relatively high ROUGE-1 score demonstrates effective identification of key content units, while ROUGE-2 confirms preservation of meaningful bigram relationships. ROUGE-L indicates strong sequence-level coherence, which is essential for accessibility applications where clarity and continuity directly influence comprehension for visually impaired users.

When segmented by document type, structured book chapters yielded slightly higher ROUGE-L scores compared to magazines and newspapers. This outcome is consistent with the linear narrative nature of books, which facilitates summarization. In contrast, magazines and newspapers contain segmented and stylistically diverse content, increasing abstraction complexity.

When analyzed by document type, performance varied slightly. Table V shows the ROUGE-L scores categorized by document type.

TABLE V ROUGE-L BY DOCUMENT TYPE

Document Type	ROUGE-L
Book	32.4
Magazine	29.7
Newspaper	30.3

These results confirm that adaptive prompt strategies improve summarization quality by tailoring compression levels to content structure rather than applying a uniform summarization approach.

D. Extractive Question Answering Performance

The extractive QA module, built using DistilBERT fine-tuned on SQuAD, demonstrated consistent answer span localization across document types. Sliding window tokenization allowed processing of long documents without truncation loss, maintaining contextual integrity.

Empirical evaluation showed that answer precision was highly correlated with extraction quality. Layout-aware extraction preserved article segmentation and heading boundaries, enabling the QA model to localize answers with higher confidence. When fallback OCR-only extraction was used, answer accuracy decreased slightly due to occasional structural ambiguity, but remained within acceptable usability thresholds.

The introduction of a confidence threshold (0.15) reduced false positives and improved perceived answer reliability. This filtering mechanism is particularly important in assistive systems where incorrect answers may mislead users.

E. End-to-End System Analysis

The integrated pipeline classification, adaptive extraction, structural reconstruction, summarization, and interactive question answering demonstrated stable and consistent behavior across heterogeneous document types. Latency measurements showed that once models were loaded at server startup, inference times remained within practical real-time limits for web-based deployment.

Resource-type awareness proved to be the central factor contributing to improved structural fidelity. Uniform OCR-only systems typically merge multi-column layouts into sequential text blocks, degrading readability and limiting downstream NLP performance. In contrast, the adaptive strategy preserved logical document segmentation prior to summarization and QA processing, resulting in improved coherence and answer localization.

Comparative evaluation against a non-adaptive baseline revealed improvements in structural accuracy, summarization coherence, and user-level interpretability. These findings validate the hypothesis that document-type awareness significantly enhances accessibility-driven document understanding systems.

V. CONCLUSION

This paper presented VisionAssist, a resource-type aware intelligent document understanding framework designed to enhance accessibility for visually impaired learners interacting with printed educational materials. Unlike conventional OCR-based systems that process heterogeneous documents uniformly, VisionAssist introduces document-type awareness as a central design principle, enabling adaptive extraction, structured reconstruction, resource-aware summarization, and interactive question answering within a unified pipeline.

Experimental evaluation demonstrates that transfer learning using MobileNetV2 achieves reliable resource-type classification with 96.57% validation accuracy, providing a robust foundation for downstream adaptive processing. Both MobileNetV2 and the baseline CNN were evaluated using identical training, validation, and test splits to ensure a fair and consistent comparison. The integration of layout-sensitive extraction strategies combining rule-based structural reconstruction for linear documents and cloud-based layout modeling for multi-column documents significantly improves structural fidelity prior to natural language processing. Furthermore, abstractive summarization using FLAN-T5 enhanced through Low-Rank Adaptation demonstrates that parameter-efficient fine-tuning can deliver competitive performance while updating only a small fraction of model parameters. The inclusion of an extractive question answering module based on DistilBERT enables meaningful and context-aware user interaction, transforming static document access into an interactive learning experience.

The results confirm that resource-type awareness substantially improves document structure preservation, summarization coherence, and answer localization compared to uniform OCR-only approaches. By coupling visual document classification with adaptive NLP strategies, VisionAssist advances the state of assistive document intelligence systems and demonstrates the practical viability of integrating computer vision, transformer-based summarization, and question answering within a scalable web-based architecture.

Overall, VisionAssist provides a real-world deployable accessibility solution that promotes independence, comprehension, and engagement for visually impaired learners. The proposed framework establishes a foundation for future research in adaptive document intelligence, where content structure and semantic understanding are jointly optimized to support inclusive and intelligent educational technologies.

VI. FUTURE WORK

Although VisionAssist demonstrates strong performance in resource-type classification, adaptive extraction, and interactive summarization, several opportunities remain for further enhancement.

First, future research will explore the integration of multimodal document understanding models such as LayoutLMv2 [12] and vision-language transformers to jointly model textual content and visual structure within a unified architecture. This may further improve structural reconstruction and reduce reliance on heuristic-based post-processing for book documents.

Second, the summarization component can be extended toward controllable and user-adaptive summarization, allowing visually impaired users to specify summary length, detail level, or focus areas dynamically. Incorporating reinforcement learning from human feedback (RLHF) or preference-based tuning may improve personalization and accessibility.

Third, the extractive question answering module may be enhanced through hybrid extractive–generative approaches. Leveraging generative QA models with retrieval augmentation could improve answer flexibility while preserving factual consistency.

Another promising direction involves multilingual expansion. Extending VisionAssist to support low-resource

and regional languages would significantly broaden accessibility impact, particularly in developing regions.

Additionally, large-scale user studies involving visually impaired learners are planned to evaluate usability, cognitive load reduction, and real-world learning effectiveness. Empirical feedback from assistive technology users will provide valuable insights for system refinement.

Finally, edge-device optimization and on-device inference techniques will be investigated to enable offline accessibility in bandwidth-limited environments. Techniques such as model quantization and knowledge distillation may reduce computational requirements while maintaining performance.

Through these future enhancements, VisionAssist aims to evolve into a fully adaptive, multilingual, and user-personalized document intelligence framework for inclusive education.

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